Building a Unified Science of Natural Behavior
with Densely Overlapping Measurement Environments [Domes]

FreeMoCap Foundation, Inc.

Written Proposal to the NSF X-Labs Initiative

1. Mission

> Develop and disseminate a new class of scientific instrument called a **Densely Overlapping Measurement Environment (Dome)** that records the complete perception–action loop of a behaving agent as a single calibrated, uncertainty-tagged measurement, and build the autonomous organization dedicated to the measurement itself to align perceptual and motor neuroscience, musculoskeletal biomechanics, mobile robotics, and embodied AI into a single convergent science of natural behavior.

A New Observable — The Sensorimotor Loop. Every agent, living or engineered, solves the same problem: sense a thin slice of the world's energy through imperfect transducers and, on that basis, push against the world to move toward its goals — information in, forces out; the brain exists to yank the bones around. We can already measure each component of this loop with extraordinary, ever-improving precision: camera and IMU motion capture for body kinematics, force plates for body–world forces, outward- and inward-facing cameras for the scene and the eyes, EEG and surface EMG for coarse neural and muscular activity in humans, and dense electrode arrays resolving individual spikes in animals. Yet the coherent whole — the shared context each tool illuminates — stays out of reach: occasionally a team stitches several threads together for a heroic integrative study, but the method rarely escapes the Methods section of a few papers spanning a single PhD.

We need a new organization that exists outside of the siloed domains of the Academia that dedicates itself to the measurement, the development and dissemination of world-class, convivial tools and novel skillsets to explore the landscape the capacity unlocks. That organization is the **FreeMoCap X-Lab** [FMC-X]. We have already changed the landscape of human-focused research by bringing high-quality motion capture to a global community. With the support of the NSF X-Labs program, we will change the face of science itself.

A New Instrument — Building the Dome. A Dome is a densely instrumented region of real space engineered to (at some aspirational extreme) record every measurable channel of the agent–environment interaction. In practice, a given Dome carries a subset of the available instrumentation, yet all Domes produce the same output. A Dome is defined by the ontology of its domain rather than its current instruments. A metrologically-grounded record of canonical models defined by their degrees of freedom (DoF), which are hydrated by measurements from whatever instruments are present. A reconstructed `Skull` is assumed to have `Eyes` whether or not an eye tracker measured them, and a `RetinalInput` or `RigidBody` denotes the same quantity whether it came from a human, a ferret, or a marmoset (See § 3 and Table 2).

The New Capability. The same **Sensor-Grounded Ontology** that lets heterogeneous Domes share one schema makes the instrument legible to both people and machines: explicit entities and relationships, and typed, grounded data over which a model can build reliable reasoning chains rather than guess at unlabeled streams. That clarity lets a Dome route its live streams and logs through a local model to assist novices, experts, and developers alike, and lets published results tag back into one shared entity–relationship structure that connects them to the wider body of science.

Unmet Needs. Building a scientific instrument is a distinct craft, not a byproduct of domain expertise. Integrated measurement is often scoped as a student side-project and abandoned when it proves to require a career of engineering rather than a semester of it. Even successes rarely outlive the trainee who built them. By Conway's Law, an enterprise of domain-named departments, journals, and grants will tend to produce only domain-scoped instruments — not the sustained metrology that turns a proof of concept into shared infrastructure. A tool defined by a measurement rather than a domain unites everyone who needs it and becomes a commons precisely because it imposes none of their goals. The measurement therefore needs its own organization, scoped to the instrument and staffed to maintain it indefinitely— what the X-Labs model exists to fund and a university appointment cannot sustain.

The Vision. The vision of this proposal represents a grand acceleration of movement the FreeMoCap Project has fostered since 2021, and there is no upper limit to our ambition. The global spread of the FreeMoCap software shows that the world has a hunger for this kind of tool, and we intend to feed it. We will build the instrument that will unlock that ability to ask an unknowably vast array of questions, and those questions will carry our tools to every corner of human curiosity. There will be a DOME in every research lab, every classroom, every Physical Therapy clinic, animation studio, and athletic facility. Field-deployable wearable DOME-MOBILE systems that provide lab-quality data of humans and animals in the remote wilderness. They will be built into the SCUBA systems of deep sea welders and into the EVA suits and HAB volumes of the ISS. Every country hospital and retirement home will have an AI-enabled, auto-calibrating, self-analyzing Gait/Posture facility that will have plain language conversations with patients about their own movements. There will be DOMEs bolted into the sides of mountains to measure the aerodynamics of hunting hawks, and onto the hulls of shipping boats to understand the mechanics of the dolphins that play in their wake. Every exotic animal enclosure in every zoo could become an automated animal behavior lab, managed through citizen science and then accelerated through auto-training backprojection pipelines.

We will build the DOME - The opposite of a space telescope. An empty volume of lenses focused inward on where the body meets the world. We will reveal a new observable and share it broadly to anyone that wants it. We will change our understanding of ourselves, and the ways we interact with the world.

2. Technology Landscape

The DOME is organized around a single, demanding measurement target: reconstructing the visual input to a behaving agent's retina precisely enough to predict the neural activity it drives. Concretely, that means estimating gaze-in-world to better than 1° — roughly the diameter of the human fovea — and resolving eye kinematics finely enough to recover retinal optic flow and its derivatives, divergence and curl ([1]; Fig. 2). Reaching that target decomposes into three measurement threads, each of which must hit its own precision and then compose into one calibrated record: the **body** that carries the eye, the **eye** itself, and the **world** it looks at. The state of the art, and the gap the DOME closes, is clearest thread by thread.

The body. Body kinematics are captured today by markerless pose estimation (e.g. FreeMoCap, riding a decade of computer-vision advances — OpenPose, DeepLabCut, MediaPipe), marker-based systems (Vicon, Qualisys, OptiTrack), and inertial (IMU) suits (Xsens), with the forces behind that motion measured by force plates (AMTI, Bertec), instrumented treadmills, and pressure insoles. Markerless capture already reaches 2–6° joint-angle accuracy against research-grade references, but in practice most users achieve far less accuracy due to errors in camera setup and calibration.

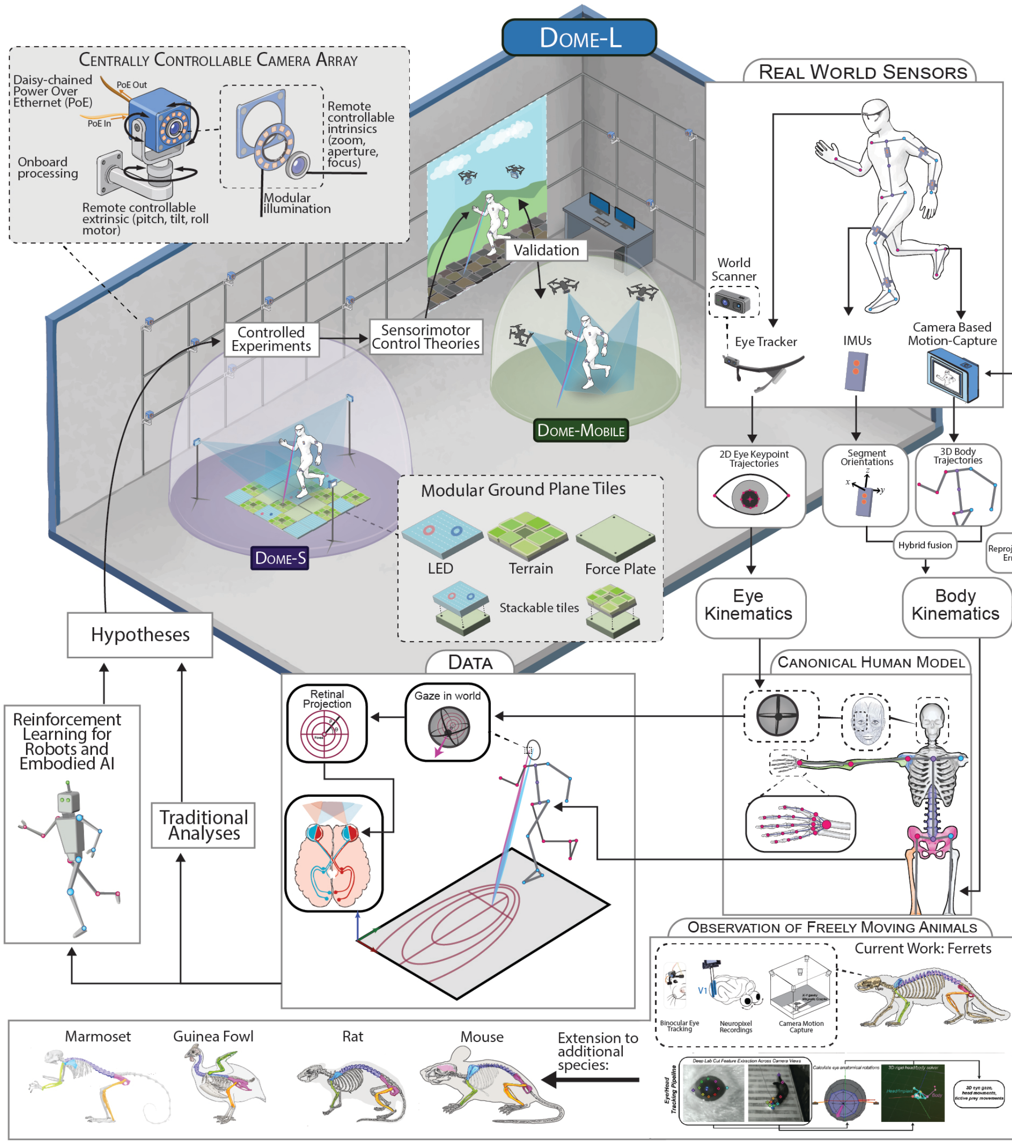

Fig. 1: A Densely Overlapping Measurement Environment (Dome). The flagship Dome-L physically contains the disseminated Dome-S and wearable Dome-Mobile forms, calibrating every variant against one reference. A centrally controllable camera array and modular force/LED/terrain floor tiles feed calibrated eye, body, world, and force streams into a single model of the sensorimotor loop — segment kinematics, gaze-in-world, and reconstructed retinal input — closing an analysis-to-hypothesis-to-experiment loop inside the same instrument that measures it.

We close that gap with a **self-calibrating, actuated camera array** that tunes its own extrinsics and intrinsics to cover a chosen capture volume from a single console, fused with IMU-suit orientation into one reproducible, uncertainty-tagged kinematic estimate — in the lab and in the field. Its mechanism and success criteria are given in § 3.

The eye. Sub-degree gaze-in-world may be reachable with commercial instruments plus FreeMoCap: the best mobile trackers reach 0.6–1.8° in the head frame (Tobii Pro Glasses 3, Pupil Labs Neon, AdHawk MindLink), and the only published end-to-end eye + IMU + world reconstruction — this team's own prior work — composes them to ±1° calibrated and ±2–3° in natural walking[2,3]. That much is careful extension of prior art. Retinal **curl**, though, is beyond every available tool, because no eye tracker measures ocular torsion. Vision science has historically ignored torsion — sensibly so, since it could not be measured and since Listing's Law specifies eye orientation from just two of its three degrees of freedom (elevation and abduction). But Listing's Law fails whenever the vestibulo-ocular reflex is active — that is, during essentially all natural behavior — and[1] implicates retinal curl as a likely control signal

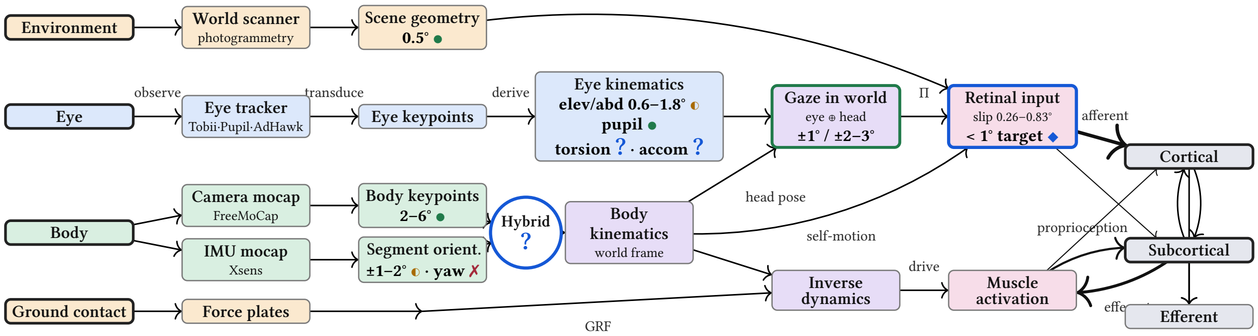

Fig. 2: Quantifying the technology landscape. Reported error sits on the **quantity measured**, not the instrument, tagged by provenance: ● open and traceable, ◐ closed vendor model (uncertainty unknown), ✗ closed and **not truth-preserving** (Xsens fills an out-of-bounds magnetometer yaw with presumptive data), ◆ Dome target, and a blue ? for what no instrument measures yet. Mobile eye trackers resolve head-frame gaze to 0.6° (Tobii Pro Glasses 3), ≈1° (AdHawk MindLink), and 1.3–1.8° (Pupil Labs Neon), but none measure ocular torsion or accommodation, and the camera–IMU fusion is itself unbuilt. Composed into gaze **in the world** through IMU segment orientation (±1–2°), markerless keypoints (2–6°,[4]), and photogrammetric scene geometry (≈3 cm ≈ 0.5°), the only published end-to-end eye + IMU + world method — this team's own prior work — reaches ±1° calibrated and ±2–3° in natural walking ([2,3]), against the sub-1° needed to predict the neural activity it drives.

for real-world locomotion, a signal the torsional axis directly shapes. Measuring that missing degree of freedom is the design target of our next-generation, dense-sensor-array eye tracker.

The world. Reconstructing what lands on the retina requires reconstructing the scene the eye looks at. Scene geometry is captured today by photogrammetry, Gaussian splatting, RGB-D, and LiDAR. The team's own photogrammetry recovers natural terrain to a repeatability near 3 cm — about 0.5° of visual angle. But these scanners are passive and stand outside the agent: none is fused into the moving observer's coordinate frame with propagated uncertainty. We mount the scanner on the agent — a dense-array world-scanner carried on the head — and extend it outdoors on the same drone-swarm nodes that carry the body cameras, so the scene is measured in the same calibrated frame as the body and eye (Fig. 1).

The gap is integration, not any single sensor. Each thread is excellent and improving on its own, but they are drifting apart rather than into a common record. Two adjacent fields are stalled for exactly this reason and would be unblocked by a calibrated, uncertainty-tagged corpus of real behavior: computer-vision pose estimation, which needs physical ground truth at scale, and legged-robot reinforcement learning, which needs real-agent recordings to close the sim-to-real gap. The Dome produces both by construction (§ 3).

3. Outcomes

The instrument. Our five-to-seven-year outcome is a single instrument — the DOME — realized in three human-scale forms and an animal-scale form that records the complete perception–action loop as one calibrated, uncertainty-tagged measurement and yields the same canonical record regardless of which sensors are present. DOME-L is the warehouse-scale flagship in Greater Boston; it drives each sub-instrument to its limit and is large enough to physically contain the smaller forms, which makes it the metrological reference for the entire network. DOME-S extends FreeMoCap's webcam capture into the disseminated lab- and classroom-scale form, and DOME-Mobile is the wearable form that carries the measurement onto natural terrain outside the lab. Because the instrument is defined by its measurement ontology rather than by any one sensor, a markerless-camera estimate and an inertial estimate describe the same kinematics and fuse into a single estimate better than either alone.

The organizing benchmark - the retinal projection. Every outcome below is a term in one uncertainty budget: reconstructing veridical retinal input — the world projected onto a moving agent's retina — to within 1° at the fovea while the agent fixates a point on the ground ahead. That target, and the current-art error sources it composes, are established in the Technology Landscape (§ 2; Fig. 2).

Body — A self-calibrating, semi-autonomous camera array. On the motor half of the loop we deliver kinematics clean enough for inverse dynamics from markerless capture. The actuated, self-calibrating camera array places daisy-chained Power-over-Ethernet cameras on programmable mounts that stay metrically calibrated as one instrument while being re-aimed and re-focused from a single console, dissolving the practical bottleneck that tends to leave large motion capture rigs frozen in place due to the high cost of manual re-calibration. This is adaptive AI-based sensing in hardware: the algorithm reconfigures the physical optics to maximize observability, running autonomous experiments over camera geometry rather than settling for a hand-placed rig. Success is measured as post-reconfiguration calibration quality and the time from sub-volume selection to calibrated capture, against a fixed-calibration reprojection baseline. Camera-derived keypoints and IMU-derived segment-orientation estimates are fused, combining noisy-but-accurate outside-in estimates with precise-but-drifting inside-out estimates under explicit per-joint uncertainty, validated as joint-torque uncertainty against force-plate ground truth and muscle-force uncertainty against EMG-informed estimates.

The starting point is already quantified: FreeMoCap recovers joint centers to under 30 mm and sagittal joint angles to 2–4° RMSE against research-grade optical motion capture during treadmill gait, from a roughly $120 six-webcam rig[4]. The actuated array and camera–IMU fusion target the residual gap to inverse-dynamics-grade kinematics — per-joint uncertainty tight enough that derived joint torques and muscle forces carry meaningful error bars.

Eye — the next generation of eye trackers. Mobile eye tracking has been stalled for at least a decade. There are more powerful arrays of multispectral optical sensors on the back of a cell phone than there are in a modern eye tracker. We will build a new kind of eye tracker that measures degrees of freedom commercial trackers omit, with spatial and temporal accuracy beyond current expectation. We will design a dense array of multi-spectral sensors (including high-resolution **stereo RGB-IR sensors, ultra-high speed event cameras, 3d scanning LiDAR, and single-pixel MEMS imagers**) to reconstruct the eye to a level of spatiotemporal precision beyond the capacity of any existing system. We will keep our unit costs low by taking advantage of the economics of the commodity-imager curve, with the goal of creating a cost-collapse for next-gen eye tracking the way FreeMoCap did for motion capture.

The new eye tracker will resolve ocular torsion from iris texture — the degree of freedom that corrupts the curl of retinal optic flow and a standing research target[1]. Realtime and post-hoc pipelines will produce microsaccade-scale resolution, and possibly unlock estimates of lens accommodation by tracking the dual-Purkinje image. Its benchmarks are sub-15-arcminute eye-in-head accuracy, 1° torsion accuracy, and the composed 1° gaze-in-world target above.

World — the scene the agent acts in. Reconstructing what the eye sees requires reconstructing the world it acts in, so a head-mounted world-scanner of stereo-RGB, structured-IR, and LiDAR produces per-frame RGB-D and, over longer horizons, a terrain mesh and a SLAM-corrected head trajectory that fixes inertial drift[3]. The same scanner package mounts on drone-swarm nodes that act as a mobile extension of the camera array, grounding DOME-Mobile's inside-out estimates outdoors where a fixed rig cannot reach, benchmarked as gaze and kinematic uncertainty against the DOME-L reference during co-recorded trials. Modular floor panels — each a force plate, terrain tile, or LED surface — close the loop by making the ground itself both sensor and experimental stimulus.

Item	In place	Phase 0 · 9–12 mo	Phase 1 · 24–36 mo	Phase 2 · variable
ORGANIZATION — OPERATIONAL DELIVERABLES				
Legal & governance	✓ FMC-F 501(c)(3)	Governance + Autonomy Plan	Board seated	
Facility (Lease)		Lease + Move-in	Office fit-out	
Full-time team	✓ Core FT	Hire · Onboard	Full staff	
Required NSF plans		Data · IP · Sec · Gov; Ph1 Plan + Budget		
IP management	✓ Prior IP	IP Mgmt Plan	License · Manage	
DOME VARIANTS				
DOME-L (Automated)		PoC / Testbed	Prototype - Validated	Operational
DOME-S (Static)	✓ FMC webcam	Validated	Operational (510(k) cert)	Operational
DOME-MOBILE	✓ Matthis/Muller papers	PoC-Breadboard	Prototype-Validated	Operational
BODY — TECHNICAL DEVELOPMENT				
Camera array		PoC-Breadboard	Prototype -> Validated	Operational
IMU↔camera fusion	✓ FMC mocap	PoC	Prototype	Validated
EYE — TECHNICAL DEVELOPMENT				
Dense Array Eye tracker		PoC - Breadboard	Prototype-Validated	Operational
WORLD — TECHNICAL DEVELOPMENT				
Dense Array World-scanner		PoC - Breadboard	Prototype-Validated	
Floor (force·LED·terrain)	✓ v1	PoC	Prototype	Validated
Drone Swarm Mocap			Concept-PoC	PoC-Validated

Table 1: Tech Readiness Ladder (TRL): Concept (2) · PoC (3) · Breadboard (4) · Prototype (5–6) · Validated (7) · Operational (8–9)

One record across species. Functionally equivalent instruments at collaborator labs yield the same canonical records as the human instrument. The animal-scale instrument already exists — the team's ferret and mouse builds integrate three-camera gaze, full-body kinematics, and world cameras in one calibrated system, with neural recordings being added now — and the human, mouse, and ferret builds are being brought to functional equivalence, extending to guinea fowl for musculoskeletal biomechanics with EMG (Monica Daley) and marmoset for primate electrophysiology (Alex Huk/Jake Yates). Success is cross-species measurement comparability under shared calibration and data-exchange protocols, so that a reconstructed `RetinalInput` or `RigidBody` denotes the same thing whether it came from a human, a ferret, or a marmoset.

What the record unlocks. Within one sensor-grounded ontology, the instrument generates its own training signal: back-projecting calibrated 3D reconstructions onto each camera's 2D view yields a reprojection-error signal that fine-tunes pose estimators on out-of-distribution movement rather than a saturated benchmark. And because a DOME export maps directly onto MuJoCo and Isaac Lab rigid-body models with uncertainty intact, DOME corpora drive sim-to-real transfer and inverse-RL recovery of the control policies underlying observed behavior — testable back in DOME-L via floor perturbation.

This self-improvement loop is already demonstrated at small scale: a generic pose estimator mistracks a transfemoral prosthesis by up to 152 mm at the ankle, but training a task-specific keypoint model on the instrument's own calibrated data drops that error to 8–14 mm[4]. What was a one-off dissertation result becomes, in the DOME, a repeatable methodology capable of driving measurement beyond the capacities of the present landscape.

4. Senior/Key Personnel

Jonathan Matthis, PhD — President / PI. Founded and maintains FreeMoCap; two decades measuring the full perception–action loop. Left a tenure-track professorship (Human Movement Neuroscience, Northeastern) because it could not support the integrated tool-building this work requires. NEI NIH K99/R00 recipient (EY028229), with a publication record spanning the full loop, from outdoor gaze-and-gait to real-world retinal-input reconstruction.

Nikki Rostellan — Chief Executive Officer. Technology-sector engineering leader and startup co-founder, recruited to run the organization so the PI can focus on the technical Mission. Staff Platform Software Engineer at Rapid7; co-founded and scaled MailLift, a venture-backed API startup. Brings the leadership and operations capacity to build FreeMoCap into an autonomous, independent X-Lab.

Endurance Idehen — Chief Technology Officer. Enterprise software architect (20+ years, engineer to Principal Architect / CTO) and FreeMoCap's CTO since 2021, essentially since founding. Concurrent CTO at Chorus Innovations; former Principal Architect at Unqork; distributed-systems and data-engineering depth. Ensures enterprise-grade architecture and the calibrated, semantically-unified data backbone the instrument requires. Represents a skillset not present in academic science.

Aaron Cherian, PhD — Chief Scientific Officer. Dissertation validated FreeMoCap against research-grade optical motion capture (Qualisys), demonstrating clinically valid kinematics from commodity hardware. Expertise in clinical tool validation, bench-to-bedside development, neuroprosthetics, and brain imaging; leads DOME-S development and clinical dissemination.

John K. Lindstedt, PhD — Chief AI Officer. Cognitive scientist and applied-AI architect (Applied AI Architect, SOLID Inc.); FreeMoCap contributor since 2024 and co-developer of SkellyBot, an AI-based teaching assistant that he and the PI built and deployed into 10 classes across 3 universities between Spring 2023 and Fall 2025. 15+ years handling sensitive human-subjects and health data under FERPA and IRB — anchors data governance and research security, and builds internal AI for development, support, and education.

Ryan Rose — Chief Financial Officer. Cooperative- and nonprofit-finance specialist and FreeMoCap's Treasurer / volunteer CFO since 2021, its longest-serving financial officer. Founder of Capital Bookkeeping Cooperative; Controller at Honest Weight Food Co-op; Adjunct Professor of Accounting, University at Albany (MS, Accounting). Runs financial operations and grant management; cooperative-governance experience fits the Foundation's open structure and the X-Labs autonomy requirements.

Karl Muller, PhD — Project Manager, DOME-Mobile. Computer-vision engineer, data scientist, and longtime PI co-author whose work maps almost one-to-one onto DOME-Mobile. Built the outdoor eye-tracking + photogrammetry + IMU pipeline the DOME generalizes[3,5]; currently Senior Data Scientist at SynMax (multi-sensor fusion, cloud MLOps). Holds essentially the complete skillset for DOME-Mobile's core instrumentation.

Michael Nguyen — Project Manager, DOME-L. Clinical biomechanics, medical-systems design, mechatronics, and multi-modal mocap-lab management; holds the skillset for large-scale DOME-L facility design and operation.

5. Team Capabilities Statement

Building the Dome demands expertise no single faculty could assemble: the work is neither pure science nor pure engineering but the sustained craft of turning one into the other, and the FreeMoCap Foundation was built to hold that combination under one roof. The leadership pairs perception–action neuroscience and clinical biomechanics with enterprise software architecture, applied-AI systems, and hardware and mechatronics engineering at wearable and facility scale, run by dedicated operations and finance leaders so the technical staff can stay on the Mission. Each member is detailed in § 4; what matters here is that they already function as a single team, having built and maintained a shipping instrument together — most for years, part-time or volunteer alongside senior roles, before any funding existed.

Governance and autonomy. The FreeMoCap X-Lab (FMC-X) operates as an independent project within the FreeMoCap Foundation, a 501(c)(3) whose leadership and mission nearly fully overlap it. Final authority rests with the President/PI, so research, partnership, and hiring decisions are made in days rather than weeks. The Foundation is already autonomous at submission — full internal control of research, funding, partnerships, IP, and hiring with no parent-institution approval required — and satisfies every condition of the NSF X-Labs

Autonomy Factor Test (§6.1.1). Phase 0 is the runway over which the remaining leaders go full-time; the PI and CSO already are.

Network, partnerships, and community. An active network of research groups already spans every domain the Dome serves — human perception-and-action researchers, robotics and prosthetics groups, visual neuroscientists working in ferret and mouse, musculoskeletal biomechanists working in guinea fowl, and primate electrophysiologists (Table 2).

Collaborator	Institution	Model
Ben Scholl	CU Anschutz	Ferret
Monica Daley	UC Irvine	Guinea fowl
Diego Fernandez	CCMH	Mouse
Alex Huk	UCLA	Marmoset, NHP
Mary Hayhoe	UT Austin	Human
Trenton Wirth	U Cincinnati	Human
Brett Fajen	RPI	Human
Kate Bonnen	Indiana Univ.	Human, NHP
Jake Yates	UC Berkeley	Human, NHP
Gabe Nelson	RAI Institute	Robotics
Chris Huibicki	FAMU-FSU	Robotics
Jonathan Hurst	Agility Robotics	Robotics
Steve Heim	Cornell	Robotics

Table 2: Collaborator network.

Under X-Lab funding, we will merge this collaborator network and the globally distributed community of FreeMoCap users to form a coherent culture organized around a shared set of tools, skills, and measurements. We will communicate design plans and roadmap priorities through structured processes that gather broad input without stalling on consensus — a Request-for-Comments process modeled on Python's PEP system.

Once a year, we will host an overlapping set of in-person events including workshops that train students and out-of-domain experts, hackathons that onboard developers, conferences that share Dome -based research, and a congress that sets organizational direction. A Community Grants Program and Developer Fund sustain the open-source network and serve as a testbed for community engagement and consensus-seeking mechanisms.

A proven model. FreeMoCap has already proven this operating model - over 15k users across 152 countries, 10k GitHub stars, and over 3,500 Discord members. It is a gift freely given, built and maintained outside academia, scoped to the measurement itself with an obsessive focus on usability and refusing to claim any research domain — and that refusal is precisely what let it become a commons rather than one lab's tool. The Dome applies the same measurement-scoped, master-built, open, full-time-maintained model to the far larger target - a scale reachable only with the dedicated, autonomous, milestone-based funding the X-Labs program exists to provide.

Intellectual property and dissemination. The IP strategy keeps the core instrumentation open while actively pursuing adoption and commercialization: an open-source core under permissive licensing for downstream use, structured to maximize adoption and impact. A full IP Management Plan is due by the end of Phase 0 under the OT agreement and will be developed against that milestone.

Adoption and dissemination will run on the engine that already built FreeMoCap. The standing collaborator network and community described above are the distribution channel: every validated Dome instance at a partner site is both a scientific node and a reference build others reproduce from a published parts list, build guide, and calibration protocol. The platform is grant-funded, so we can sell hardware at very low margin to reach a price point that ensures maximum penetration across the widest possible slice of society. The open-source core invites downstream products, services, and integrations to form around one shared measurement standard, built without black-boxes or IP-induced cones of silence, so the X-Lab's reach is counted not only in units sold but in the independent instruments, companies, and published results that adopt the shared philosophy organized around the love of measurement, and the expansive drive to see more and more fully than we ever could before.

References

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